Example 1.9

Mention the property for the following equations:

(i) $(-45)+(-12)=-57$ (ii) $(-15)+7=(7)+(-15)$

(iii) $-10+3=-7$ (iv) $(-7)+(-5)=(-5)+(-7)$

(v) $(-7)+[(-4)+(-3)]=[(-7)+(-4)]+(-3)$ (vi) $0+(-7245)=-7245$

Solution

(i) Closure Property (ii) Commutative Property

(iii) Closure Property (iv) Commutative Property

(v) Associative Property (vi) Additive Identity

Exercise 1.1

1. Fill in the blanks

(i) $(-30)+____=60$ (ii) $(-5)+____=-100$

(iii) $(-52)+(-52)=____$ (iv) $____+(-22)=0$

(v) $____+(-70)=70$ (vi) $20+80+____=0$

(vii) $75+(-25)=____$ (viii) $171+_____=0$

(ix) $[(-3)+(-12)]+(-77)=_____+[(-12)+(-77)]$

(x) $(-42)+[____+(-23)]=[____+15]+____$.

2. Say true or false.

(i) The additive inverse of (–32) is (–32)

(ii) $(-90)+(-30)=60$

(iii) $(-125)+25=-100$

3. Add the following

(i) 8 and –12 using number line (ii) (–3) and (–5) using number line

(iii) $(-100)+(-10)$ (iv) $20+(-72)$

(v) $82+(-75)$ (vi) $-48+(-15)$

(vii) $-225+(-63)$

4. Thenmalar appeared for competitive exam which has negative scoring of 1 mark for each incorrect answer. In paper I she answered 25 questions incorrectly and in paper II, 13 questions incorrectly. Find the total reduction of marks.

5. In a quiz competition, Team A scored +30,–20, 0 and team B scored –20, 0, +30 in three successive rounds. Which team will win? Can we say that we can add integers in any order?

6. Are $(11+7)+10$ and $11+(7+10)$ equal? Mention the property.

7. Find 5 pairs of integers that add up to 2.

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